

Note 5.7 Orthogonal Polynomials

Orthogonal Sequences

Definition of Orthogonal Polynomials

For a sequence of polynomials, if the **inner product** of each pair of them are zero, the sequence is a **sequence of orthogonal polynomials**. Moreover, if its elements are **unit**, the sequence is a sequence of **orthonormal polynomials**.

Theorem 5.7.1

For a sequence of orthogonal polynomials $p_0, p_1, \dots$,

- $p_0, \dots, p_{n-1}$ form a basis for P_n .
- p_n is orthogonal to **every polynomials of degree less than n** . ($p_n \in P^T$)

And we can know that

$$p = \sum_{i=0}^{n-1} \frac{\langle p, p_i \rangle}{\langle p_i, p_i \rangle} p_i = \sum_{i=0}^{n-1} \frac{\langle f, p_i \rangle}{\langle p_i, p_i \rangle} p_i$$

where p is the **best least approximation to f by the elements of P_n** .

Theorem 5.7.2

For a sequence of orthogonal polynomials $p_0, \dots$, let a_i denote the **lead coefficient of p_i** for each i and $p_{-1}(x)$ be zero polynomials, then we have

$$\alpha_{n+1}p_{n+1}(x) = (x - \beta_{n+1})p_n(x) - \alpha_n\gamma_n p_{n-1}(x) \quad (n \geq 0)$$

where $\alpha_0 = \gamma_0 = 1$ and

$$\alpha_n = \frac{a_{n-1}}{a_n}, \beta_n = \frac{\langle p_{n-1}, xp_{n-1} \rangle}{\langle p_{n-1}, p_{n-1} \rangle}, \gamma_n = \frac{\langle p_n, p_n \rangle}{\langle p_{n-1}, p_{n-1} \rangle}$$

The theorem can use **three-term recursion relation** to find the elements in the sequence.

We are **free to choose a_n** since any **nonzero multiple of the orthogonal polynomials are also orthogonal polynomials**. Since we can choose all $a_i = 1$ such that

$$p_{n+1}(x) = (x - \beta_{n+1})p_n(x) - \gamma_n p_{n-1}(x) \quad (n \geq 0)$$

Theorem 5.7.3

For a sequence of orthogonal polynomials with the inner product

$$\langle p, q \rangle = \int_a^b p(x)q(x)w(x) dx$$

, then the zero points of $p_n(x)$ are all real and distinct and lie in the interval (a, b) .